KAIZEN MATHS

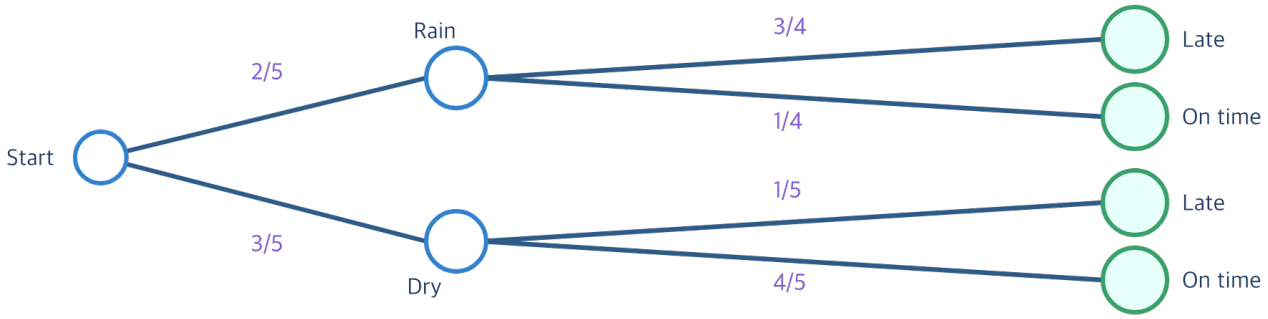
Lesson starter

Tree Diagrams and Conditional Probability

Reverse Probability

Name _____

SA1



The tree shows a situation where P(Rain) = $\frac{2}{5}$, P(Late | Rain) = $\frac{3}{4}$, and P(Late | Dry) = $\frac{1}{5}$. Find P(Late).

STA-3EBB-L3 · SA

Show your working

KAIZEN MATHS

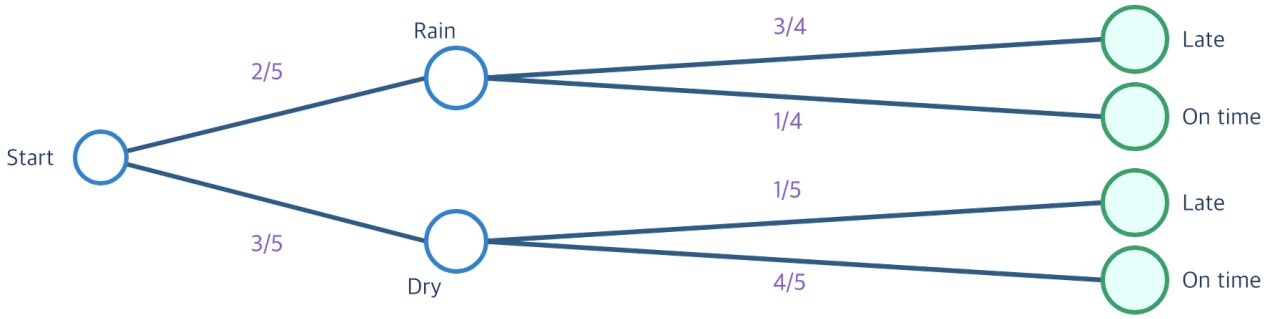
Lesson starter

Tree Diagrams and Conditional Probability

Reverse Probability

Name _____

SB1



The tree gives P(Rain) = $\frac{2}{5}$, P(Late | Rain) = $\frac{3}{4}$, and P(Late | Dry) = $\frac{1}{5}$. Given that Late happened, find P(Rain | Late).

STA-3EBB-L3 · SB

Show your working

KAIZEN MATHS

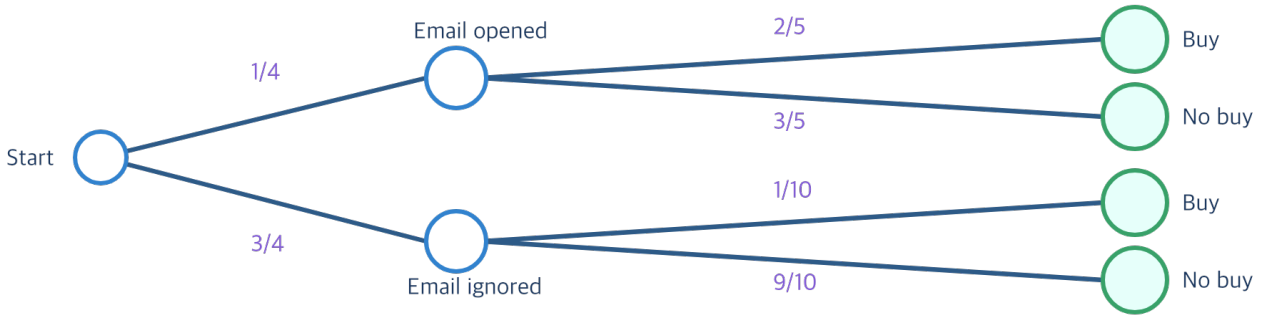
Lesson starter

Tree Diagrams and Conditional Probability

Reverse Probability

Name _____

SC1



The tree shows a situation where P(Email opened) = $\frac{1}{4}$, P(Buy | Email opened) = $\frac{2}{5}$, and P(Buy | Email ignored) = $\frac{1}{10}$. Find P(Buy).

STA-3EBB-L3 · SC

Show your working

KAIZEN MATHS

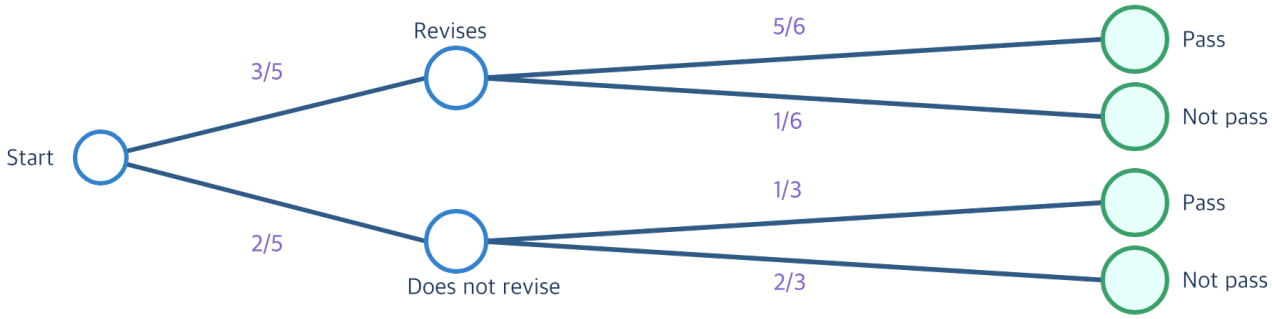
Lesson starter

Tree Diagrams and Conditional Probability

Reverse Probability

Name _____

SD1



The tree gives P(Revises) = $\frac{3}{5}$, P(Pass | Revises) = $\frac{5}{6}$, and P(Pass | Does not revise) = $\frac{1}{3}$. Given that Pass happened, find P(Revises | Pass).

STA-3EBB-L3 · SD

Show your working

KAIZEN MATHS

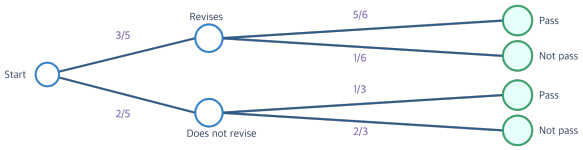
Exit ticket

Tree Diagrams and Conditional Probability

Reverse Probability

Name _____

EA1



The tree shows a situation where P(Revises) = $\frac{3}{5}$, P(Pass | Revises) = $\frac{5}{6}$, and P(Pass | Does not revise) = $\frac{1}{3}$. Find P(Pass).

Show your working

KAIZEN MATHS

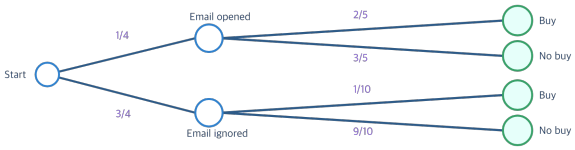
Exit ticket

Tree Diagrams and Conditional Probability

Reverse Probability

Name _____

EB1



The tree gives P(Email opened) = $\frac{1}{4}$, P(Buy | Email opened) = $\frac{2}{5}$, and P(Buy | Email ignored) = $\frac{1}{10}$. Given that Buy happened, find P(Email opened | Buy).

Show your working

KAIZEN MATHS

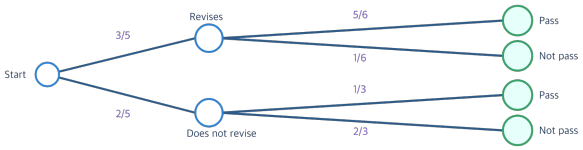
Exit ticket

Tree Diagrams and Conditional Probability

Reverse Probability

Name _____

EA1



The tree shows a situation where P(Revises) = $\frac{3}{5}$, P(Pass | Revises) = $\frac{5}{6}$, and P(Pass | Does not revise) = $\frac{1}{3}$. Find P(Pass).

STA-3EBB-L3 · EA

Show your working

KAIZEN MATHS

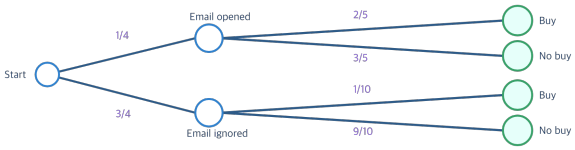
Exit ticket

Tree Diagrams and Conditional Probability

Reverse Probability

Name _____

EB1



The tree gives P(Email opened) = $\frac{1}{4}$, P(Buy | Email opened) = $\frac{2}{5}$, and P(Buy | Email ignored) = $\frac{1}{10}$. Given that Buy happened, find P(Email opened | Buy).

STA-3EBB-L3 · EB

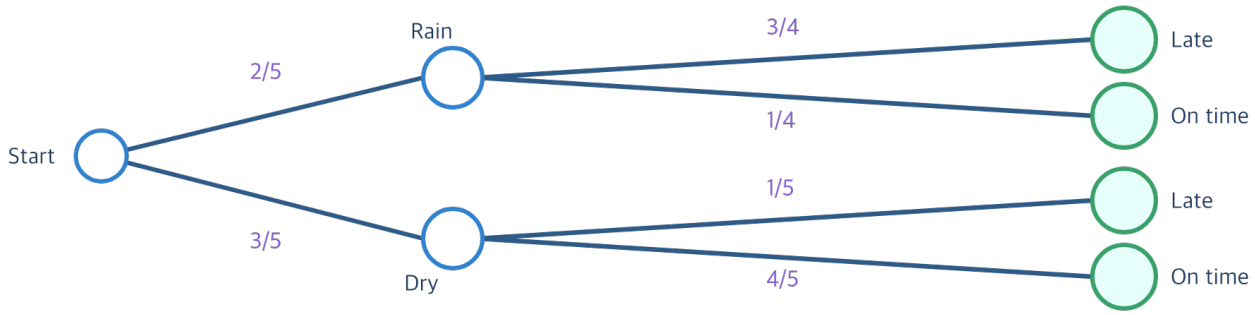
Show your working

Teacher answers & working

Tree Diagrams and Conditional Probability · Reverse Probability

STA-3EBB-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



The tree shows a situation where P(Rain) = $\frac{2}{5}$, P(Late | Rain) = $\frac{3}{4}$, and P(Late | Dry) = $\frac{1}{5}$. Find P(Late).

Answer:

$\frac{21}{50}$

Working:

To get Late, add every path that ends in Late.

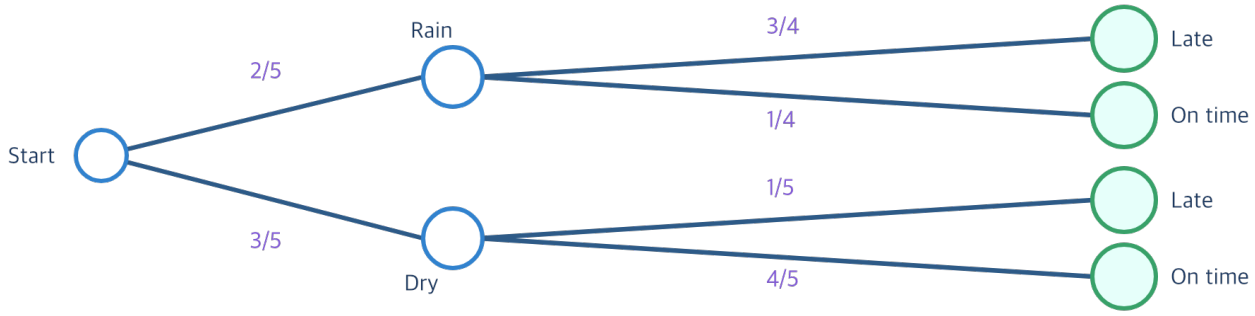
Rain and Late: $\frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$

Dry and Late: $\frac{3}{5} \times \frac{1}{5} = \frac{3}{25}$

P(Late) = $\frac{3}{10} + \frac{3}{25} = \frac{21}{50}$

$\frac{21}{50} = 42.0\%$

SB1



The tree gives P(Rain) = $\frac{2}{5}$, P(Late | Rain) = $\frac{3}{4}$, and P(Late | Dry) = $\frac{1}{5}$. Given that Late happened, find P(Rain | Late).

Answer:

$\frac{5}{7}$

Working:

Reverse conditional probability asks for the matching path divided by all paths in the condition.

Matching path: P(Rain and Late) = $\frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$

All Late paths: $\frac{3}{10} + \frac{3}{25} = \frac{21}{50}$

P(Rain | Late) = $\frac{3}{10} \div \frac{21}{50} = \frac{5}{7}$

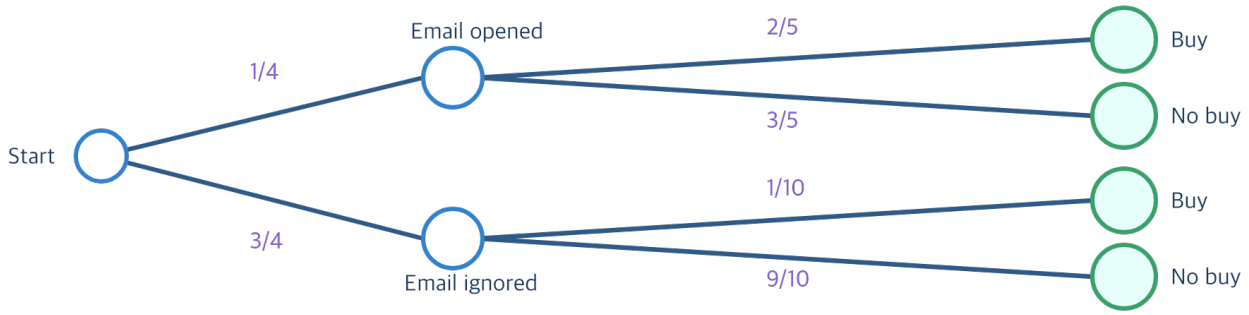
$\frac{5}{7} = 71.4\%$

Teacher answers & working

Tree Diagrams and Conditional Probability · Reverse Probability

STA-3EBB-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



The tree shows a situation where P(Email opened) = $\frac{1}{4}$, P(Buy | Email opened) = $\frac{2}{5}$, and P(Buy | Email ignored) = $\frac{1}{10}$. Find P(Buy).

Answer:

$\frac{7}{40}$

Working:

To get Buy, add every path that ends in Buy.

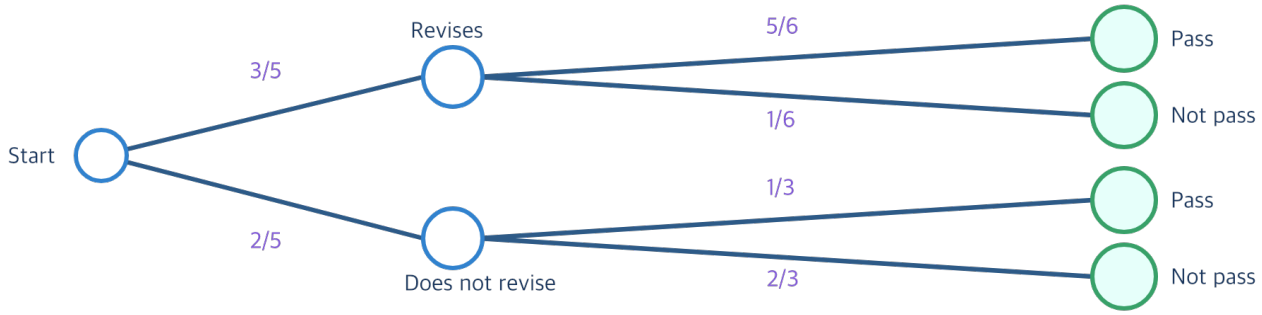
Email opened and Buy: $\frac{1}{4} \times \frac{2}{5} = \frac{1}{10}$

Email ignored and Buy: $\frac{3}{4} \times \frac{1}{10} = \frac{3}{40}$

$P(\text{Buy}) = \frac{1}{10} + \frac{3}{40} = \frac{7}{40}$

$\frac{7}{40} = 17.5\%$

SD1



The tree gives P(Revises) = $\frac{3}{5}$, P(Pass | Revises) = $\frac{5}{6}$, and P(Pass | Does not revise) = $\frac{1}{3}$. Given that Pass happened, find P(Revises | Pass).

Answer:

$\frac{15}{19}$

Working:

Reverse conditional probability asks for the matching path divided by all paths in the condition.

Matching path: P(Revises and Pass) = $\frac{3}{5} \times \frac{5}{6} = \frac{1}{2}$

All Pass paths: $\frac{1}{2} + \frac{2}{15} = \frac{19}{30}$

P(Revises | Pass) = $\frac{1}{2} \div \frac{19}{30} = \frac{15}{19}$

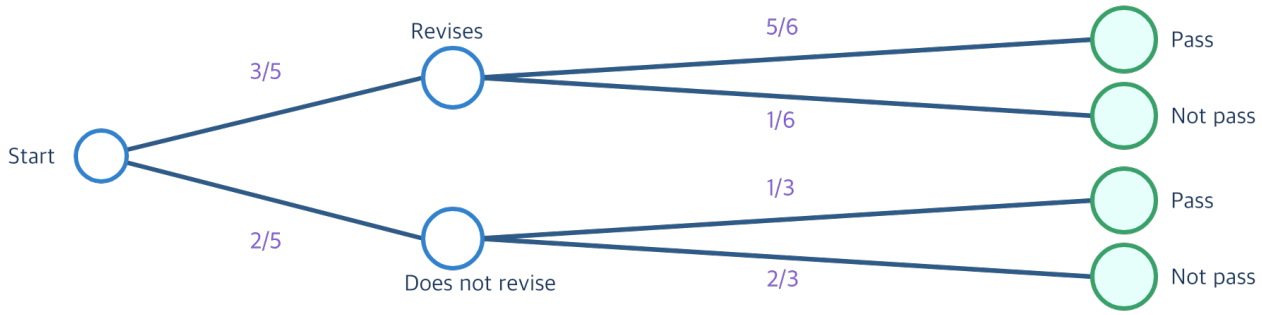
$\frac{15}{19} = 78.9\%$

Teacher answers & working

Tree Diagrams and Conditional Probability · Reverse Probability

STA-3EBB-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



The tree shows a situation where P(Revises) = $\frac{3}{5}$, P(Pass | Revises) = $\frac{5}{6}$, and P(Pass | Does not revise) = $\frac{1}{3}$. Find P(Pass).

Answer:

$\frac{19}{30}$

Working:

To get Pass, add every path that ends in Pass.

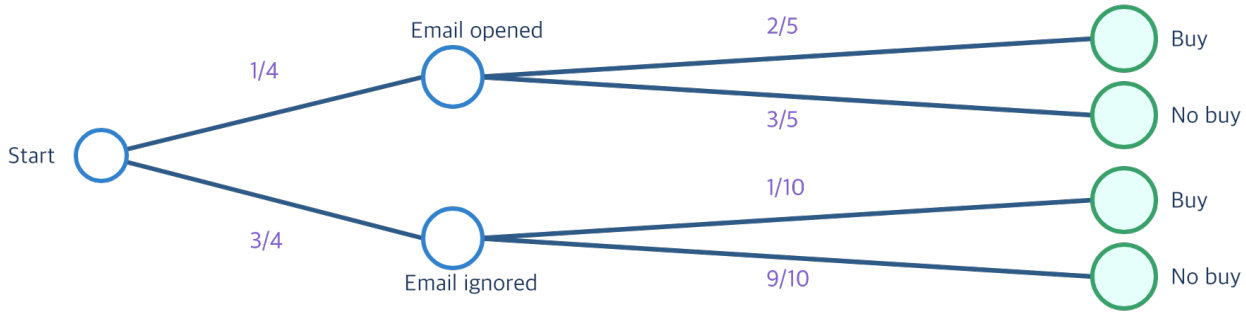
Revises and Pass: $\frac{3}{5} \times \frac{5}{6} = \frac{1}{2}$

Does not revise and Pass: $\frac{2}{5} \times \frac{1}{3} = \frac{2}{15}$

P(Pass) = $\frac{1}{2} + \frac{2}{15} = \frac{19}{30}$

$\frac{19}{30} = 63.3\%$

EB1



The tree gives P(Email opened) = $\frac{1}{4}$, P(Buy | Email opened) = $\frac{2}{5}$, and P(Buy | Email ignored) = $\frac{1}{10}$. Given that Buy happened, find P(Email opened | Buy).

Answer:

$\frac{4}{7}$

Working:

Reverse conditional probability asks for the matching path divided by all paths in the condition.

Matching path: P(Email opened and Buy) = $\frac{1}{4} \times \frac{2}{5} = \frac{1}{10}$

All Buy paths: $\frac{1}{10} + \frac{3}{40} = \frac{7}{40}$

P(Email opened | Buy) = $\frac{1}{10} \div \frac{7}{40} = \frac{4}{7}$

$\frac{4}{7} = 57.1\%$